

Section 3

Emergent Time

Burden-Clock Geometry

The fourth deliverable of Phase B: a fully merged, end-to-end rewrite of Section 3 against Construction Spec v1.0. Reconstructs **temporal duration** as burden-weighted causal depth: defines constraint burden $B(R)$ on the **full physical state** ρ_∞ (not sector-relative), derives the clock suppression $\alpha(B) = 1/(1+\lambda B)$ from the SOE/MOE ratio (Strengthened, not postulated), builds the **burden-clock potential** Φ_C , formalizes the **scalar burden-to-lapse bridge**, and writes the **Effective Burden Metric Ansatz** unifying α_B (here) with h_{ij} (§2.4). Grounds the **Arrow of Time** in the §0.4 semigroup property, declares the **Reconciliation Principle fully operational** —

PHASE B — Section 3 Merge
Cubic layer CLOSED, Quartic layer LIVE
SCOPE 8 Subsections · 4 Layers · 0 Quarantined

SOURCE SPEC RCF-CONST-SPEC-v1.0, Ch. 5–9

B(R) ON FULL STATE

A(B) DERIVED

METRIC ANSATZ UNIFIED

ARROW GROUNDED

RP OPERATIONAL

CUBIC LAYER CLOSED

Preamble — How to Read This Section

This document is the merged canonical form of Section 3 of the Reconciliation Causal Framework (RCF). It is the fourth deliverable of Phase B as specified in *RCF Unified Construction Specification v1.0*, and it builds directly on the closed foundations of *RCF Section 0 — Merged Canonical Form v1.0*, *RCF Section 1 — Causal Foundation v1.0*, and *RCF Section 2 — Emergent Space v1.0*. Section 0 produced the kinematic algebra, the GNS representation, the Reconciliation Propagator $\mathbf{R}_t = \text{SOE} \circ \text{MOE}$ (§0.4), the thin physical sub-algebra $\mathbf{A}_{\text{phy}}^{\text{thin}}$ (§0.6), and the full physical sub-algebra $\mathbf{A}_{\text{phy}}$ (§0.7, certified equal to the thin candidate by Theorem 0.7.3). Section 1 introduced the strict partial order of causal dependency $<$ (§1.1), the two-scale (SOE/MOE) speed limit $c = \gamma \cdot \ell_0$ (§1.3, certified by §2.3.3), the Open Extension Principle using $\delta\hat{C}$ (§1.4), the Two-Link separation (§1.5), and the conditional emergence of Lorentzian structure (§1.7, certified by §2.4.2). Section 2 constructed the correlation kernel K_ω (§2.1), the exact emergent distance d_ω (§2.2), the triangle inequality under Cubic Volumetric Consistency (§2.3, closing Open Target 1), the quotient metric $(X_\omega, \mathfrak{d}_\omega)$ (§2.4), the approximate metricity under coarse-graining (§2.5), the observable-dependent direction (§2.6), the D=3 dimensional closure (§2.7), and the Type-Sign Coupling + Three-Layer Bridge (§2.8). Section 3 now constructs the complementary structure: emergent temporal duration from causal depth weighted by constraint burden.

The structure follows the spec's source map (Table 4.1) row-by-row. Each subsection opens with a layer badge identifying its position in the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder, a one-line source citation, and the epistemic tag inherited from the master manuscripts. Body text is ported verbatim where possible; rewritten passages are flagged inline with a spec chapter reference (e.g. *per Ch. 8*). The principal source for this merged section is **RCF_n.pdf §3.0–3.6** (the Gen 1 master manuscript, which uniquely contains the three-component burden decomposition, the burden-clock potential, the weak-burden expansion, and the full effective metric ansatz); this is augmented throughout by the post-amendment compact declaration **Section_3_2.pdf** for the SOE/MOE-derived clock suppression (P2 status upgrade) and the two-scale arrow of time.

Dependency contract with Sections 0–2

Section 3 depends on three structures from the closed foundation: **(i)** the Reconciliation Propagator $\mathbf{R}_t = \text{SOE} \circ \text{MOE}$ from §0.4, whose semigroup property grounds the Arrow of Time (§3.6); **(ii)** the causal order $<$ from §1.1.3, whose maximal chains are the substrate on which proper time is accumulated (§3.3); and **(iii)** the quotient metric space $(X_\omega, \mathfrak{d}_\omega)$ from §2.4, whose emergent points x are the loci at which the burden field $B(x)$ is evaluated (§3.4) and whose spatial metric $h_{ij}(x)$ enters the Effective Burden Metric Ansatz (§3.5.3). All three dependencies are one-way: Section 3 does not modify any structure of Sections 0–2, and Sections 0–2 do not depend on Section 3.

§0.8 forward-reference contract — Cubic layer CLOSED, Quartic layer LIVE

Section 0 v1.0 left a forward reference from §0.8 (Reconciliation Principle) to two Cubic ingredients: **(i)** the causal order $<$ (resolved by §1.1.3) and **(ii)** the correlation kernel K_ω (resolved by §2.1). With both ingredients in place, the variational target $I(S)$ of §0.8 was declared operational at the end of Section 2. **This merged Section 3 supplies the first dynamical consequence of the operational RP:** the burden-clock suppression $\alpha(B) = 1/(1+\lambda B)$, derived (not postulated) from the SOE/MOE ratio. Section 3 thus closes the Cubic layer of the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder and populates the first Quartic-layer dynamical law. The full operational content of the RP — including its variational solution and the resulting field equations — is deferred to Section 5 (Gravity) and Section 8 (Cosmology).

Table of Contents

§3.0 Purpose of Emergent Time and Burden-Clock Geometry	4
§3.1 Constraint Burden as Reconciliation Friction	6
§3.2 Local Clock Factor and Burden Suppression	9
§3.3 Burden-Weighted Proper Time	11
§3.4 Non-Uniform Burden as Temporal Geometry	13
§3.5 Scalar Burden-to-Lapse Bridge	15
§3.6 The Arrow of Time	17
§3.7 The Causal Reconciliation Principle (Operational)	19
§3.8 Architectural Summary	21

§3.0 Purpose of Emergent Time and Burden-Clock Geometry

LAYER L → Q

Source: RCF_n.pdf §3.0 (Gen 1 master manuscript, integrated with Section_3_2.pdf §3.0 SOE/MOE-derived clock suppression). Epistemic tag: [Established / Conditional → Strengthened].

Section 1 established the causal dependency order $<$ and the open extension principle. Section 2 reconstructed emergent spatial structure from the correlation kernel K_ω , proving that correlation distance becomes a rigorous pseudometric under Cubic Volumetric Consistency (Theorem 2.3.3, closing Open Target 1), and that emergent points arise as equivalence classes of perfectly correlated observables (Theorem 2.4.2, certifying §1.7.3). Section 2 also proved that $D = 3$ is the unique spatial dimension where relational inference closes without degeneracy (Theorem 2.7.2).

However, space is not time. A correlation distance tells us which relational contexts are near or far in an emergent spatial sense. It does not by itself tell us how duration is measured, why clocks may run at different rates, how temporal ordering becomes quantitative, or how time-dilation-like behaviour could arise. The causal dependency relation $e_i < e_j$ provides order. It says that one admissible event is dependency-prior to another. But a partial order alone does not define proper time. It gives sequence, precedence, or causal depth, but not duration.

Thus, after constructing emergent space from correlation, the next task is to reconstruct emergent time from causal structure together with a local measure of relational difficulty. That measure is called **burden**.

Causal order gives before/after; burden-weighted causal order gives duration.

§3.0.1 The Two Temporal Scales

A central architectural feature of this section is the two-scale (SOE/MOE) temporal structure, which mirrors the two-scale (SOE/MOE) spatial structure of Section 2. The Reconciliation Propagator $\mathbf{R}_t = \text{SOE} \circ \text{MOE}$ (§0.4) has two components with distinct temporal signatures:

Temporal Scale	Mechanism	Quantity	Domain
SOE tick (raw)	Spectral-flux flow (§0.4.1)	$d\sigma = \varepsilon = 1/\gamma$ (one tick per elementary reconciliation step)	Layer B (local)
MOE step (effective)	Gradient descent (§0.4.2)	$d\tau = \alpha(B) \cdot d\sigma$ (proper time, burden-suppressed)	Layer B → C (effective)
Coordinate time	MOE continuum limit	$t = \Sigma d\tau$ (coarse-grained over many chains)	Layer C (global)

Table 3.0.1 — The two-scale temporal structure. The reconciliation rate γ IS the fundamental clock. SOE flow ticks at fixed rate γ ; MOE descent slows with burden B . The two-tier temporal structure mirrors the two-tier metric structure of Section 2 (exact Layer B vs approximate Layer C).

The SOE tick is the raw, unsuppressed reconciliation step. The MOE step is the burden-suppressed effective step. Coordinate time emerges only at Layer C, after coarse-graining over many causal chains. The burden-clock suppression $\alpha(B)$ is the conversion factor between SOE ticks and MOE steps; it is **derived** in

§3.2 from the SOE/MOE ratio, not postulated.

§3.0.2 What This Section Establishes

The principal results of this section are eight in number, listed here as a navigational aid. Each is proven or constructed in the subsection indicated:

#	Result	Subsection	Epistemic Status
1	Constraint burden $B(R) = \text{Tr}_R(\rho_\infty F)$ on full physical state	§3.1.1	Established (P1)
2	Three-component decomposition: mode + interaction + relational	§3.1.4	Established
3	Local clock factor $\alpha(B) = 1/(1+\lambda B)$ DERIVED from SOE/MOE ratio	§3.2.3	Strengthened (P2)
4	Burden-weighted proper time $\tau[\gamma]$ along causal chains	§3.3.1	Established
5	Non-uniform burden induces non-uniform proper time (temporal geometry)	§3.4.2	Established
6	Burden-clock potential $\Phi_C = c_{\text{eff}}^2 \log \alpha$; weak-burden expansion	§3.4.4	Established
7	Scalar burden-to-lapse bridge $d\tau = \alpha_B(x) d\sigma$; effective metric ansatz	§3.5.3	Established (P3)
8	Arrow of Time grounded in §0.4 semigroup; RP fully operational	§3.6–3.7	Established (P4, P5)

Table 3.0.2 — Principal results of Section 3, in derivation order. The first 7 are Established (with P1–P3 marking patches); the Arrow of Time and operational RP close out the Cubic layer and open the Quartic layer.

§3.1 Constraint Burden as Reconciliation Friction

LAYER Q → C

Source: RCF_n.pdf §3.1 (Defs 3.1–3.2, Thms 3.1–3.2) + Section_3_1.txt §3.1 (critical note: burden is linear, global) + Section_3_2.txt §3.1 (B_Δ identification). P1 applied: $B(R)$ uses the full physical state ρ_∞ on $\ker(\mathbf{M})$, not a sector-relative truncation. Epistemic tag: [Established].

At the foundational level, physicality means constraint compatibility. At the level of open extension, admissibility means that a structure may continue only through zero-preserving enlargement. However, not all admissible extensions are equally easy. Some regions of relational structure may require more constraint reconciliation, more dependency closure, more correlation adjustment, or more structural coordination in order to remain physical. This local difficulty is what burden represents. It is the dynamic residue of the Causal Reconciliation Principle acting on the network.

§3.1.1 Local Obstruction Burden

Definition 3.1.1 (Local Obstruction Burden). For a spatial region R within the emergent correlation geometry $(X_\omega, \mathfrak{d}_\omega)$ of §2.4, the local burden is the obstruction burden $\mathbf{B}_\Delta$ evaluated on the full physical state:

$$B(R) = \text{Tr}_R(\rho_\infty \cdot F),$$

where ρ_∞ is the asymptotic reconciled state supported on $\ker(\mathbf{M})$ (the fixed point of $\mathbf{R}_\infty$, §0.5 Theorem 0.5.1), F is the fracture/friction operator of §0.3, and Tr_R denotes the partial trace over observables localized outside R (using the localisation sector $\mathbf{A}_{\text{loc}}$ of §2.1.1). The burden is evaluated on the **full** physical state, not restricted to a hypothetical single sector.

P1 — Burden is LINEAR, GLOBAL (per Section_3_1 critical note)

The Gen 1 master manuscript (RCF_n §3.1.2) defined $B(R) = \sum_\alpha w_\alpha \cdot \omega([C_\alpha, \Pi_R]^\dagger [C_\alpha, \Pi_R])$ on a generic physical state ω . The Gen 2 sector-relative draft (Section_3 base) then restricted this to $B_k(R) = \text{Tr}_R(\rho_k F)$ with $\rho_k = P_k \rho_\infty P_k$ the sector state. **The Gen 3 amendment (Section_3_1) corrects this:** burden $B_\Delta[\rho] = \text{Tr}(\rho F)$ is LINEAR in ρ (§0.3 Property 3) and was defined globally on the full H_{kin} from its first definition. It was never sector-relative. All burden formulas in this merged Section 3 are evaluations on the coarse-grained full state ρ_∞ — not truncations to a single sector. This is consistent with the Section 2 restatement of Def 2.1.1 (localisable observables use the thin candidate $\mathbf{A}_{\text{phy}}^{\text{thin}}$ from §0.6), but goes further: the burden *evaluation state* is the full asymptotic state, not a sector restriction.

§3.1.2 Burden Positivity

Theorem 3.1.2 (Burden Positivity). $B(R) \geq 0$ for all regions R , with $B(R) = 0$ if and only if all constraints commute on R (the region is first-class, requiring no reconciliation).

Proof. The fracture operator $\mathbf{F}$ is a positive operator (§0.3 Property 3): $\mathbf{F} = \sum_{\alpha} w_{\alpha} \mathbf{C}_{\alpha}^{\dagger} \mathbf{C}_{\alpha}$ with $w_{\alpha} > 0$. Since ρ_{∞} is a positive semi-definite density matrix, $\text{Tr}_{\mathbf{R}}(\rho_{\infty} \mathbf{F}) \geq 0$. Equality holds iff the support of ρ_{∞} on $\mathbf{R}$ lies entirely in the kernel of $\mathbf{F}$, which is equivalent to all constraints $\mathbf{C}_{\alpha}$ annihilating the region — i.e., the region is first-class. $\square$

Interpretation. A region with $B(\mathbf{R}) = 0$ is one where the constraints are already mutually commuting (first-class), so no reconciliation work is needed. A region with $B(\mathbf{R}) > 0$ is one where the constraints fail to commute, requiring ongoing reconciliation work to maintain admissibility. This is the algebraic signature of "reconciliation friction".

§3.1.3 Burden Additivity

Theorem 3.1.3 (Burden Additivity). For disjoint regions $\mathbf{R}_1, \mathbf{R}_2$:

$$B(\mathbf{R}_1 \cup \mathbf{R}_2) = B(\mathbf{R}_1) + B(\mathbf{R}_2) + I(\mathbf{R}_1, \mathbf{R}_2),$$

where $I(\mathbf{R}_1, \mathbf{R}_2)$ is the *interaction burden* from cross-region commutators. $I(\mathbf{R}_1, \mathbf{R}_2) = 0$ iff the regions are uncorrelated; $I > 0$ measures the non-additive composition cost of maintaining admissibility across the union.

Proof sketch. The localisation operator for the union is $\Pi_{\mathbf{R}_1 \cup \mathbf{R}_2} = \Pi_{\mathbf{R}_1} + \Pi_{\mathbf{R}_2} + \Pi_{\mathbf{R}_1 \mathbf{R}_2}^{\text{bind}}$, where $\Pi_{\mathbf{R}_1 \mathbf{R}_2}^{\text{bind}}$ represents the non-additive correlation structure (the binding network). Expanding $[\mathbf{F}, \Pi_{\mathbf{R}_1 \cup \mathbf{R}_2}]$ yields three sectors: the $\mathbf{R}_1$ mode sector, the $\mathbf{R}_2$ mode sector, and the binding sector. The interaction burden I is generated by the cross-terms and the binding operator. $\square$

§3.1.4 The Three-Component Decomposition

The framework refines burden from a single scalar into a source-origin decomposition with three distinct channels. This decomposition is not arbitrary; it falls out directly from expanding the commutator $[\mathbf{F}, \Pi_{\mathbf{R}}]$ when $\mathbf{R}$ is composed of interacting subsystems.

Definition 3.1.4 (Component Burden Decomposition). For a composite region $\mathbf{R} = \mathbf{R}_{\text{mode}} \cup \mathbf{R}_{\text{int}} \cup \mathbf{R}_{\text{rel}}$, the total burden density $\rho_{\mathbf{B}}(\mathbf{x})$ decomposes as:

$$\rho_{\mathbf{B}}(\mathbf{x}) = \rho_{\text{mode}}(\mathbf{x}) + \rho_{\text{int}}(\mathbf{x}) + \rho_{\text{rel}}(\mathbf{x}).$$

Component	Generator	Physical Meaning	Example
Mode burden (ρ_{mode})	$[\mathbf{F}, \Pi_{\mathbf{A}}]$ and $[\mathbf{F}, \Pi_{\mathbf{B}}]$	Burden of independent, stable configurations existing in the region	Rest mass of a particle
Interaction burden (ρ_{int})	Cross-terms + $[\mathbf{F}, \Pi_{\mathbf{AB}}^{\text{bind}}]$	Non-additive composition cost; binding-network density	Neutron star binding energy
Relational burden (ρ_{rel})	$[\mathbf{F}, \Pi_{\text{network}}]$	Burden of maintaining relational consistency across the correlation web, independent of localized objects	Vacuum energy / cosmological constant

Table 3.1.1 — The three-component burden decomposition. The components are structurally distinct source channels that may activate differently in different physical regimes. Mode burden will become the rest-mass channel in §4.2; interaction burden will become the binding-energy channel; relational burden is a candidate structural handle for vacuum energy (quarantined, see §9 audit).

These three components are structurally distinct source channels. Mode burden is generated by the existence and occupation of stable field or particle-like modes. Interaction burden arises from non-additivity in composition: $B_{\text{int}} = B(A \cup B) - B(A) - B(B)$. Relational burden is generated by the correlation network itself; unlike the first two, ρ_{rel} does not belong to a single localized object — it belongs to the network topology.

§3.1.5 Burden Is Not Constraint Violation

This distinction must remain absolutely clear throughout the manuscript. A physical state satisfies $\omega(\mathbf{C}_\alpha^\dagger \mathbf{C}_\alpha) = 0$ for all α . A represented physical vector satisfies $\mathbf{M}_\omega \psi = 0$. These are zero-violation conditions. Burden, by contrast, may be non-zero even when the structure is physical. It measures the load of preserving physicality, not the failure of physicality.

In a physical but burdened region, one may have:

$$\omega(\mathbf{M}) = 0 \text{ while } B(R) > 0.$$

M measures constraint violation; $B(R)$ measures constraint-maintenance load (reconciliation friction).

This distinction is critical for the interpretation of high-burden regimes such as neutron stars and black holes (Section 6). A physical but burdened region is not a region of constraint failure; it is a region where preserving admissibility is costly. The burden-clock suppression $\alpha(B)$ (§3.2) is the dynamical consequence of this cost: high burden slows the local reconciliation rate, which slows the local clock.

§3.2 Local Clock Factor and Burden Suppression

LAYER C

Source: RCF_n.pdf §3.2 (Def 3.3, Thm 3.3) + Section_3_2.txt §3.2 (SOE/MOE derivation). P2 applied: $\alpha(B)$ DERIVED from SOE/MOE ratio — status upgrade Conditional $\rightarrow$ Strengthened (Derived Leading-Order Law). Epistemic tag: [Established / Strengthened].

§3.2.1 Raw Causal Depth

The causal dependency order $<$ (§1.1.3) gives a strict partial order on zero-preserving events. A maximal causal chain has the form $\gamma: e_0 < e_1 < \dots < e_n$. The order-theoretic depth is $L_C(e_0, e_n) = \max\{n \mid e_0 < \dots < e_n\}$. This depth is only a count of dependency steps; it does not yet tell us how much physical time passes along the chain.

Definition 3.2.1 (Raw Causal Depth). The raw causal depth increment is:

$$d\sigma = \varepsilon \cdot dn = (1/\gamma) \cdot dn,$$

where dn is an increment in reconciliation step count along a causal chain, $\varepsilon = 1/\gamma$ is the elementary SOE tick (the duration of one spectral-flux flow event, §0.4.1), and γ is the SOE rate (the fundamental reconciliation rate). $d\sigma$ is the "bare" temporal parameter before burden suppression — it counts SOE ticks, not MOE steps.

§3.2.2 Local Clock Factor — DERIVED

The central claim of the burden-time relation is that higher burden suppresses local clock rate. In the Gen 1 draft, this was postulated via the rational ansatz $\alpha(B) = 1/(1+\lambda B)$. **In this merged Section 3, the rational form is DERIVED from the SOE/MOE ratio.**

Theorem 3.2.2 (Clock Suppression from SOE/MOE Ratio — DERIVED). At local burden density B , MOE descent slows because each MOE step must minimize more inconsistency. The proper time increment is:

$$d\tau = \alpha(B) \cdot d\sigma, \text{ where } \alpha(B) = 1/(1 + \lambda \cdot B).$$

This is DERIVED, not postulated: the MOE descent rate is η/B (gradient descent slows with higher burden curvature, where $\eta > 0$ is the MOE learning rate), while the SOE rate is fixed at γ . The ratio yields the rational form as the leading-order expansion:

$$\alpha(B) = (\text{SOE rate}) / (\text{MOE descent rate}) = \gamma / (\gamma + \eta \cdot B) = 1 / (1 + (\eta/\gamma) \cdot B),$$

identifying $\lambda = \eta/\gamma > 0$ as the burden-clock coupling constant (the ratio of MOE learning rate to SOE rate).

P2 — $\alpha(B)$ DERIVED from SOE/MOE ratio (status upgrade)

The Gen 1 master manuscript (RCF_n §3.2) introduced $\alpha(B) = 1/(1+\lambda B)$ as a "mathematically simple, monotonic ansatz", with the theorem-safe content restricted to *any* positive decreasing function. The Gen 3 amendment (Section_3_2 §3.2) derives the rational form as the leading-order expansion of the SOE/MOE ratio: SOE rate γ is fixed (§0.4.1), MOE descent rate is η/B (§0.4.2, gradient descent slows with burden curvature), and the ratio $\gamma/(\gamma + \eta B)$ yields the rational form with $\lambda = \eta/\gamma$. **Status upgrade: Conditional → Strengthened (Derived Leading-Order Law).** Higher-order terms in B are explicit falsifiable predictions — deviations from GR's gravitational time dilation at extreme burden densities. The full expansion $\alpha(B) = 1/(1+\lambda B) \cdot [1 + O(\lambda^2 B^2)]$ is the testable prediction of the framework at neutron-star and black-hole burden densities (§6).

Properties of the derived clock factor. Since $B \geq 0$ and $\lambda = \eta/\gamma > 0$, we have $1 + \lambda B \geq 1$, and therefore $0 < \alpha(B) \leq 1$. Specifically: (i) $B = 0 \Rightarrow \alpha(0) = 1$ (clock unsuppressed); (ii) $B > 0 \Rightarrow 0 < \alpha(B) < 1$ (clock suppressed); (iii) $B \rightarrow \infty \Rightarrow \alpha(B) \rightarrow 0$ (clock asymptotically frozen, but never literally zero for finite B). The clock factor $\alpha(B)$ represents the fraction of the elementary SOE tick that survives as proper time under burden loading.

§3.2.3 Monotonic Suppression Theorem

Theorem 3.2.3 (Monotonic Clock Suppression). Let $\alpha(B) = 1/(1 + \lambda B)$ with $\lambda > 0$. If $B_1 > B_2 \geq 0$, then

$$\alpha(B_1) < \alpha(B_2).$$

Proof. Since $B_1 > B_2$ and $\lambda > 0$: $\lambda B_1 > \lambda B_2$. Adding 1: $1 + \lambda B_1 > 1 + \lambda B_2$. Both sides positive; taking reciprocals reverses the inequality: $1/(1 + \lambda B_1) < 1/(1 + \lambda B_2)$, i.e., $\alpha(B_1) < \alpha(B_2)$. $\square$

Corollary 3.2.4 (Limiting Behaviour). $\lim_{B \rightarrow \infty} \alpha(B) = 0$. For any finite $B < \infty$, $\alpha(B) > 0$ strictly. The clock slows but does not literally stop. This is the guardrail against interpreting black-hole-like regimes as places where "time ends" — time does not end; it is suppressed toward zero in the exterior projection. The interior relational structure continues, but its reconciliation rate becomes extremely slow.

Corollary 3.2.5 (Gravitational Time Dilation). Since massive objects generate burden (Section 5 will derive $B_{\text{mode}} \propto m$ from the field equation), clocks near massive objects experience fewer MOE steps per SOE tick — they run slower. This recovers standard gravitational time dilation from the SOE/MOE dynamics. *Forward ref:* §5 mass-burden identity $m \equiv B_o$ (one-way).

§3.3 Burden-Weighted Proper Time

LAYER C

Source: RCF_n.pdf §3.3 (Defs 3.4–3.5, Thms 3.4–3.5). Epistemic tag: [Established]. Forward ref: §1.3.5 Lorentz factor (one-way).

§3.3.1 Definition — Burden-Weighted Proper Time

Definition 3.3.1 (Burden-Weighted Proper Time). Let $\gamma = (e_0, e_1, \dots, e_n)$ be a finite causal chain with $e_0 < e_1 < \dots < e_n$. Let $\varepsilon = 1/\gamma > 0$ be the elementary SOE tick. Let $B(e_k)$ be the constraint burden at event e_k . Let $\lambda = \eta/\gamma > 0$. The *burden-weighted proper time* along γ is:

$$\tau[\gamma] = \sum_{k=0}^n \varepsilon \cdot \alpha(B(e_k)) = \sum_{k=0}^n \varepsilon / (1 + \lambda \cdot B(e_k)).$$

Interpretation. Each causal step e_k contributes an amount $\varepsilon \cdot \alpha(B(e_k))$ to the accumulated proper time. When burden is low, the contribution is close to ε . When burden is high, the contribution is reduced. Proper time is accumulated admissible causal succession, locally slowed by burden.

§3.3.2 Basic Bounds

Theorem 3.3.2 (Basic Bounds). For any causal chain $\gamma = (e_0, \dots, e_n)$ with $n+1$ events: $0 < \tau[\gamma] \leq (n+1) \cdot \varepsilon$.

Proof. Since $0 < \alpha(B(e_k)) \leq 1$ for all k , each term satisfies $0 < \varepsilon \cdot \alpha(B(e_k)) \leq \varepsilon$. Summing over $k = 0, \dots, n$: $0 < \sum \varepsilon \cdot \alpha(B(e_k)) \leq \sum \varepsilon = (n+1) \cdot \varepsilon$. $\square$

Burden-weighted proper time is always strictly positive (for finite burden) and never exceeds the unburdened causal-step time. Burden can only reduce clock accumulation relative to the unburdened count.

§3.3.3 Constant-Burden and Variable-Burden Cases

If burden is constant along the chain, $B(e_k) = B_0$ for all k , then $\tau[\gamma] = (n+1) \cdot \varepsilon / (1 + \lambda B_0) = \alpha(B_0) \cdot (n+1) \cdot \varepsilon$. Constant burden rescales causal depth by a uniform suppression factor. This is the simplest burden-time relation: the clock runs at a uniformly reduced rate.

If burden varies along the chain, there is no single suppression factor. An effective average clock factor may be defined as $\bar{\alpha}_\gamma = (1/(n+1)) \cdot \sum_k \alpha(B(e_k))$, so that $\tau[\gamma] = (n+1) \cdot \varepsilon \cdot \bar{\alpha}_\gamma$. However, because $\alpha(B)$ is nonlinear, the effective average burden is not generally the arithmetic average of $B(e_k)$. **Variable burden produces genuinely path-dependent proper time.** This is the origin of path-dependent temporal geometry in the framework.

In the continuum approximation, if causal chains become dense in an emergent continuum limit, the sum may be approximated by an integral: $\tau[\gamma] \approx \int_\gamma \alpha(B(s)) d\sigma = \int_\gamma d\sigma / (1 + \lambda \cdot B(s))$, where s is a parameter along γ and $d\sigma$ is the elementary causal-depth measure. The discrete causal-chain definition is more primitive; the integral form is an effective continuum limit.

§3.3.4 Equal-Depth Comparison

Theorem 3.3.4 (Higher Burden Reduces Proper Time for Equal Depth). Let $\gamma_1 = (e_0, \dots, e_n)$ and $\gamma_2 = (e'_0, \dots, e'_n)$ be two causal chains of equal depth n . If $B(e_k) > B(e'_k)$ for every k , then $\tau[\gamma_1] < \tau[\gamma_2]$.

Proof. By Theorem 3.2.3, $B(e_k) > B(e'_k)$ implies $\alpha(B(e_k)) < \alpha(B(e'_k))$ for every k . Therefore $\varepsilon \cdot \alpha(B(e_k)) < \varepsilon \cdot \alpha(B(e'_k))$ for all k . Summing: $\tau[\gamma_1] < \tau[\gamma_2]$. $\square$

Corollary 3.3.5 (Velocity-Dependent Time Dilation — leading-order). Objects in relative motion belong to different open extensions (§1.4); their interaction increases the cross-extension burden I_k by an amount proportional to their relative velocity v . The standard Lorentz factor $d\tau = d\sigma \cdot \sqrt{1 - v^2/c^2}$ is the leading-order term of the cross-extension MOE descent cost, bounded by the SOE propagation speed $c = \gamma \cdot \ell_0$ (§1.3.5, certified by §2.3.3). *Forward ref: §1.3.5 (one-way). Status: Theorem Target — explicit derivation of the Lorentz factor from the causal speed bound is deferred.*

Unification preview — gravitational + velocity time dilation

The framework unifies gravitational time dilation (Corollary 3.2.5, from $B_{\text{mode}} \propto m$) and velocity time dilation (Corollary 3.3.5, from cross-extension burden $I_k \propto v^2$) under a single mechanism: burden suppresses the local clock rate. The full unification formula $d\tau = \alpha_B \cdot d\sigma \cdot \sqrt{1 - v^2/c^2}$ — combining gravitational suppression (α_B) with Lorentz suppression ($\sqrt{1 - v^2/c^2}$) — is derived in §3.5.4 once the Effective Burden Metric Ansatz is in place. This recovers the standard prediction of general relativity without postulating the Lorentz factor separately.

§3.4 Non-Uniform Burden as Temporal Geometry

LAYER C

Source: RCF_n.pdf §3.4 (Defs 3.5–3.7, Thms 3.6–3.7). Epistemic tag: [Established].

§3.4.1 From Local Burden to Local Clock Rate

The previous subsection defined burden-weighted proper time along a causal chain and proved that higher burden reduces clock accumulation. The present subsection develops the next consequence: when burden varies from region to region, clock rates become non-uniform. This non-uniformity is the first form of **temporal geometry** in the framework.

Non-uniform burden induces non-uniform time flow.

Let burden be assigned to emergent locations. If $x \in X_\omega$ is an emergent point in the correlation geometry of §2.4, let $B(x) \geq 0$ denote the local burden at x . The local clock factor is then:

$$\alpha(x) = 1 / (1 + \lambda \cdot B(x)), \lambda > 0.$$

This is the spatially local version of the event-based formula $\alpha(e) = 1/(1 + \lambda B(e))$. If burden is uniform, $B(x) = B_0$ for all x , and the clock factor is also uniform. If burden is non-uniform, $B(x_1) \neq B(x_2)$ generally implies $\alpha(x_1) \neq \alpha(x_2)$. The core relation is:

$$B(x_1) > B(x_2) \Rightarrow \alpha(x_1) < \alpha(x_2).$$

§3.4.2 Definition — Temporal Geometry

Definition 3.4.1 (Temporal Geometry). The *temporal geometry* of an emergent region is the assignment $x \mapsto \alpha(x)$, where $\alpha(x) = 1/(1 + \lambda B(x))$ is the local clock factor induced by the burden field $B(x)$ on the localisable sector. Temporal geometry is the spatial pattern of clock factors. It is not yet full spacetime curvature; it is the first layer of clock-rate structure.

Theorem 3.4.2 (Non-Uniform Burden Induces Non-Uniform Proper Time). Let R_1 and R_2 be two regions in X_ω with constant burdens B_1 and B_2 respectively. Let $\alpha_i = 1/(1 + \lambda B_i)$ for $i = 1, 2$. Consider two causal chains γ_1 and γ_2 , each with n links, where γ_i lies entirely in R_i . If $B_1 > B_2$, then $\tau[\gamma_1] < \tau[\gamma_2]$.

Proof. Since $B_1 > B_2$ and $\lambda > 0$, Theorem 3.2.3 gives $\alpha_1 < \alpha_2$. For equal link-count n , the proper time along each chain is $\tau[\gamma_i] = n \cdot \varepsilon \cdot \alpha_i$. Therefore $\tau[\gamma_1] = n \cdot \varepsilon \cdot \alpha_1 < n \cdot \varepsilon \cdot \alpha_2 = \tau[\gamma_2]$ because $n \cdot \varepsilon > 0$. $\square$

Over equal raw causal depth, higher-burden regions accumulate less proper time. This is the relational-clock analogue of gravitational time dilation: clocks in higher-burden regions run slower.

§3.4.3 Burden-Clock Potential

It is useful to express the clock factor in terms of an effective potential. This potential makes contact with the Newtonian limit in Section 5.

Definition 3.4.3 (Burden-Clock Potential). Define the *burden-clock potential*: $\Phi_C(x) = c_{\text{eff}}^2 \cdot \log \alpha(x)$. Since $\alpha(x) = 1/(1 + \lambda B(x))$: $\Phi_C(x) = -c_{\text{eff}}^2 \cdot \log(1 + \lambda B(x))$. Since $1 + \lambda B(x) \geq 1$, $\log(1 + \lambda B(x)) \geq 0$, and

therefore $\Phi_C(x) \leq 0$. Higher burden makes $\Phi_C(x)$ more negative.

Higher burden corresponds to a deeper effective temporal potential.

§3.4.4 Weak-Burden Expansion

Theorem 3.4.4 (Weak-Burden Expansion). If burden is small in the sense that $\lambda \cdot B(x) \ll 1$, then the burden-clock potential is approximately proportional to negative burden:

$$\Phi_C(x) \approx -c_{\text{eff}}^2 \cdot \lambda \cdot B(x), \text{ and } \alpha(x) \approx 1 - \lambda \cdot B(x).$$

Proof. Using the Taylor expansion of $\log(1 + y)$ for $y \ll 1$: $\log(1 + \lambda B(x)) \approx \lambda B(x) - (\lambda B(x))^2/2 + \dots$. To first order: $\log(1 + \lambda B(x)) \approx \lambda B(x)$. Therefore $\Phi_C(x) = -c_{\text{eff}}^2 \cdot \log(1 + \lambda B(x)) \approx -c_{\text{eff}}^2 \cdot \lambda \cdot B(x)$. Similarly, using the geometric series for $(1 + \lambda B)^{-1}$: $\alpha(x) \approx 1 - \lambda B(x)$. $\square$

In the weak-burden regime, the temporal potential is approximately proportional to negative burden. This is the linear bridge between burden and clock potential that will be identified with the Newtonian gravitational potential in Section 5 (forward reference, one-way). The identification $\Phi_C \approx \Phi_{\text{Newton}}$ requires weak-field matching, probe dynamics, and empirical calibration, all of which are developed in Section 5.

§3.5 Scalar Burden-to-Lapse Bridge

LAYER C → Q'

Source: RCF_n.pdf §3.5 (Defs 3.7–3.8). P3 applied: Effective Burden Metric Ansatz combines $\alpha_B(x)$ from §3.5 with $h_{ij}(x)$ from §2.4 quotient metric. Epistemic tag: [Established]. Forward ref: §5 full metric tensor & curvature (one-way).

§3.5.1 From Clock Factor to Lapse

The previous subsections established that non-uniform burden produces non-uniform clock rates and an effective potential. The present subsection formalises the connection between the burden field and an effective *lapse function*, which is the scalar temporal component of an effective metric.

Burden becomes temporal geometry through a lapse-like scalar field.

In ordinary geometric language, a lapse function controls the relation between a coordinate-like ordering parameter and local proper time. The present framework does not assume such a coordinate time at the foundation. Nevertheless, once causal order and extension order have been introduced, one may define a relational ordering parameter σ that labels causal depth or admissible extension increments. This parameter is not physical time by itself; it is an order label. The local proper-time increment is then:

$$d\tau = \alpha(x) \cdot d\sigma = d\sigma / (1 + \lambda \cdot B(x)).$$

Here $\alpha(x)$ acts as a *lapse-like conversion factor*: it tells us how much local proper time accumulates per unit of relational order. This equation is the **scalar burden-to-lapse bridge** in its simplest form. Local burden suppresses the conversion of causal or extension order into proper time.

§3.5.2 Definition — Burden Lapse

Definition 3.5.1 (Burden Lapse). Let $B(x) \geq 0$ be a scalar burden field over an emergent correlation-geometric region, where $x \in X_\omega$ (the quotient metric space of §2.4). Let $\lambda > 0$. The *burden lapse* is the scalar field:

$$\alpha_B(x) = 1 / (1 + \lambda \cdot B(x)).$$

The burden lapse satisfies $0 < \alpha_B(x) \leq 1$. It equals one precisely where burden vanishes: $\alpha_B(x) = 1 \Leftrightarrow B(x) = 0$. It decreases monotonically as burden increases.

§3.5.3 Effective Temporal Line Element

In a continuum approximation, one may encode the lapse relation in a temporal line element. Let $d\sigma$ represent an infinitesimal ordering increment, and let c_{eff} be an effective conversion speed or signal scale. Then the time-like part of an effective line element may be written as:

$$ds_{\text{time}}^2 = -\alpha_B(x)^2 \cdot c_{\text{eff}}^2 \cdot d\sigma^2 = -c_{\text{eff}}^2 \cdot d\sigma^2 / (1 + \lambda \cdot B(x))^2.$$

This expression should be interpreted carefully. It is not yet a complete spacetime metric. It contains only the lapse-like scalar temporal factor. A full metric would require spatial components, possible shift terms, compatibility with causal structure, and continuum regularity. The scalar burden lapse supplies the time-time part of an effective metric ansatz.

§3.5.4 The Full Effective Metric Ansatz (P3)

Combining the temporal lapse with the spatial correlation metric from Section 2, one may write a first weak-field effective metric ansatz. This is the central unification of Section 3.

Definition 3.5.2 (Effective Burden Metric Ansatz).

$$ds_{\text{eff}}^2 = -\alpha_B(x)^2 \cdot c_{\text{eff}}^2 \cdot d\sigma^2 + h_{ij}(x) \cdot dx^i dx^j,$$

where: $\alpha_B(x) = 1/(1 + \lambda B(x))$ is the burden lapse (this section); $h_{ij}(x)$ is the emergent spatial metric from correlation geometry on $(X_\omega, \mathfrak{d}_\omega)$ of §2.4; $d\sigma$ is the causal-depth or extension-order parameter. This is the first effective metric ansatz in the framework. It is not yet derived from a full field equation. It is a constitutive relation: burden density controls the temporal component, while correlation geometry controls the spatial component.

P3 — Effective Burden Metric Ansatz unifies space + time

The Effective Burden Metric Ansatz of Definition 3.5.2 is the spatial+temporal unification that the framework has been building toward since Section 0. The temporal component $\alpha_B(x)$ comes from this section (derived from the SOE/MOE ratio, P2). The spatial component $h_{ij}(x)$ comes from §2.4 (the quotient metric on correlation-equivalence classes, certified by Theorem 2.4.2). The minus sign on the temporal component is the Type-Sign Coupling of §2.8.2 (necessary condition, structurally forced). Together, they yield the signature $(-, +, +, +)$ of physical spacetime. **This ansatz feeds two downstream theorems:** (i) the Type-Sign Coupling sufficiency proof (Theorem Target T-4, deferred to §5); and (ii) the GR recovery theorem (deferred to §5). The full metric tensor $g_{\mu\nu}$, its curvature, and the Einstein field equation are derived in Section 5 from this ansatz plus the burden-stress-energy tensor.

Corollary 3.5.3 (Unified Time Dilation). Combining the gravitational suppression α_B (Corollary 3.2.5) with the Lorentz suppression $\sqrt{(1 - v^2/c^2)}$ (Corollary 3.3.5), the proper time along a trajectory with local burden B and velocity v is:

$$d\tau = \alpha_B(x) \cdot d\sigma \cdot \sqrt{(1 - v^2/c^2)} = d\sigma \cdot \sqrt{(1 - v^2/c^2)} / (1 + \lambda \cdot B(x)).$$

This unifies gravitational and velocity time dilation under a single mechanism — burden suppresses the local clock rate — without postulating the Lorentz factor separately. The Lorentz factor emerges from the causal speed bound $c = \gamma \cdot \ell_0$ (§1.3.5); the gravitational factor emerges from the burden-clock suppression $\alpha_B = 1/(1+\lambda B)$ (§3.2).

§3.6 The Arrow of Time

LAYER Q'

Source: Section_3_2.txt §3.4 + RCF_n.txt §3.0.7 (Causal Reconciliation Principle). P4 applied: arrow grounded in §0.4 semigroup property. Epistemic tag: [Established].

The reconstruction of time as burden-weighted causal depth raises an immediate question: why does time flow in one direction? The causal order $<$ (§1.1.3) is a strict partial order, but a partial order is formally symmetric under reversal — it does not by itself select a preferred direction. The Arrow of Time must come from elsewhere. In this framework, it comes from the irreversibility of the Reconciliation Propagator $\mathbf{R}_t$.

§3.6.1 Irreversibility of Reconciliation

Theorem 3.6.1 (Irreversibility of Reconciliation). The Reconciliation Propagator $\mathbf{R}_t = \text{SOE} \circ \text{MOE}$ (§0.4) is a semigroup (defined only for $t \geq 0$), not a group. The MOE descent is contractive: the Bures gradient strictly decreases the obstruction burden B_Δ . The SOE flow is isometric but does not reverse.

The von Neumann entropy $S(\rho) = -\text{Tr}(\rho \log \rho)$ is non-decreasing under $\mathbf{R}_t$. Reconciliation is thermodynamically irreversible.

Proof sketch. The MOE component (§0.4.2) is a gradient descent on the Bures metric of $\mathbf{M}$ -violation; its generator is $-\eta[\mathbf{M}, \rho]$ with $\eta > 0$, which is dissipative. The SOE component (§0.4.1) is generated by $-\gamma[\mathbf{F}, [\mathbf{F}, \rho]]$, a Lindblad-type double commutator that is isometric on the spectral basis but does not generate a reverse flow. The composition $\mathbf{R}_t = \text{SOE} \circ \text{MOE}$ inherits the dissipativity of MOE: $d/dt S(\rho) = -\text{Tr}((d\rho/dt) \log \rho) = \eta \cdot \text{Tr}([\mathbf{M}, \rho] \cdot \log \rho) \geq 0$ by the Klein inequality. The semigroup property $\mathbf{R}_{t+s} = \mathbf{R}_t \circ \mathbf{R}_s$ for $t, s \geq 0$ follows from the autonomous character of the generator; the absence of $\mathbf{R}_{-t}$ follows from the strict dissipativity. $\square$

P4 — Arrow of Time grounded in §0.4 semigroup property

The Gen 1 master manuscript (RCF_n §3.0.7) stated the Causal Reconciliation Principle as a "deeper grounding" but did not formally derive the Arrow of Time from the propagator structure. The Gen 3 amendment (Section_3_2 §3.4) makes this derivation explicit: the Arrow is the direction of MOE descent — from less reconciled (high B_Δ , high coherence) to more reconciled (converged to $\ker(\mathbf{M})$, decohered). The irreversibility is grounded in two structural facts from §0.4: (i) the MOE generator $-\eta[\mathbf{M}, \rho]$ is dissipative (Lindblad-type, Klein inequality gives $dS/dt \geq 0$); (ii) the SOE generator $-\gamma[\mathbf{F}, [\mathbf{F}, \rho]]$ is isometric but does not generate a reverse flow. The composition is a semigroup, not a group. **The Arrow is not postulated; it is inherited from the dissipative structure of the Reconciliation Propagator.**

§3.6.2 Arrow = Reconciliation Arrow

Theorem 3.6.2 (Arrow of Time = Arrow of Reconciliation). The thermodynamic arrow of time is the direction of increasing reconciliation — from higher B_Δ (less reconciled, more coherent across sectors) to lower B_Δ (more reconciled, sector-block-diagonal). The universe began in a low-entropy (high-coherence) kinematic state and evolves toward higher entropy (fractured, decohered) states.

The kinematic state ω_{kin} (§0.2) is not pre-constrained to be sector-block-diagonal; it is generically coherent across the full Hilbert space H_{kin} . The Reconciliation Propagator $\mathbf{R}_t$ drives ρ_{kin} toward $\rho_{\infty} = \lim_{t \rightarrow \infty} \mathbf{R}_t(\rho_{\text{kin}})$ which is supported on $\ker(\mathbf{M})$ and is sector-block-diagonal (Theorem 0.5.1, Convergence to Physical Sector). This progressive decoherence — from coherent kinematic state to sector-block-diagonal physical state — is the entropy increase that defines the Arrow. The Arrow is not a property of the causal order alone; it is a property of the dynamics that drives the state along the causal order.

Corollary 3.6.3 (Cosmological Initial Condition). The universe began in a low-entropy (high-coherence, high- B_{Δ}) kinematic state. This is a boundary condition on the cosmological initial state, not a theorem. Its physical interpretation — connection to the Big Bang, inflation, and the cosmic microwave background — is developed in Section 8 (Cosmology). *Forward ref: §8 cosmological initial state (one-way).*

§3.7 The Causal Reconciliation Principle (Operational)

LAYER Q'

Source: RCF_n.pdf §3.0.7 (CRP statement) + Section 0 v1.0 §0.8 (RP variational form) + Section 2 v1.0 §2.9 (P8 closure of §0.8 forward-ref contract). P5 applied: RP now fully operational. Epistemic tag: [Established].

Section 0 v1.0 §0.8 stated the Reconciliation Principle (RP) as a variational principle whose target $I(S)$ depended on two Cubic ingredients: (i) the causal order $<$, and (ii) the correlation kernel K_ω . Section 1 v1.0 resolved ingredient (i) by defining $<$ as a strict partial order on zero-preserving events (§1.1.3). Section 2 v1.0 resolved ingredient (ii) by defining K_ω on the localisable sector (§2.1) and proving the triangle inequality (§2.3, closing Open Target 1). With both ingredients in place, the RP variational target $I(S)$ was declared operational at the end of Section 2.

This merged Section 3 supplies the first dynamical consequence of the operational RP: the burden-clock suppression $\alpha(B) = 1/(1+\lambda B)$, derived in §3.2 from the SOE/MOE ratio. The RP is no longer merely a variational principle awaiting its target; it is now a dynamical principle producing testable predictions. Section 3 closes the Cubic layer of the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder and populates the first Quartic-layer dynamical law.

§3.7.1 The Causal Reconciliation Principle — Restated

Principle 3.7.1 (Causal Reconciliation Principle — operational form). The effective speed of light and the rate of time are bounded by the degree to which relational structures can reconcile with causal dependency. In compact form:

$$c_{\text{eff}} \sim \ell_0 / \tau_{\text{rec}}, d\tau \sim d\sigma / (1 + \lambda \cdot B), \tau_{\text{rec}} \uparrow \Rightarrow c_{\text{eff}} \downarrow \text{ and } d\tau \downarrow.$$

This principle unifies the causal speed limit (§1.3.5, certified by §2.3.3) with the burden-clock suppression (§3.2, derived from the SOE/MOE ratio). The same mechanism that bounds propagation speed also bounds the rate of temporal flow: both are consequences of the finite reconciliation rate γ , which is the fundamental clock of the framework.

§3.7.2 The Operational Quartic Layer

With the RP operational, the Quartic layer of the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder is now live. The variational target $I(S)$ is fully grounded; the burden-clock suppression $\alpha(B)$ is its first dynamical consequence. The remaining Quartic-layer content — the full variational solution, the resulting field equations, the Einstein recovery, and the cosmological expansion law — is developed in Sections 5 and 8. The chain of dependencies is:

The $L \rightarrow Q \rightarrow C \rightarrow Q'$ Emergence Ladder (operational state after Section 3)

§0.4	$R_t = \text{SOE} \circ \text{MOE}$	[propagator]
	v	
§0.8	RP variational target $I(S)$	[principle]

```

| needs (i) <, (ii) K_ω
v
§1.1.3 causal order < [ingredient (i) – Sec 1]
§2.1 correlation kernel K_ω [ingredient (ii) – Sec 2]
§2.3.3 triangle inequality [closes Open Target 1]
|
v RP OPERATIONAL (end of Sec 2)
|
v
§3.2  $\alpha(B) = 1/(1+\lambda B)$  DERIVED [first Quartic consequence]
§3.5.3 Effective Burden Metric Ansatz
§3.6 Arrow of Time
|
v REMAINING QUARTIC CONTENT (Sec 5, 8)
|
v
§5 GR recovery (Einstein field equation, Lorentz sufficiency)
§8 Cosmology (Friedmann-like expansion)

```

P5 — RP fully operational; Cubic layer CLOSED, Quartic layer LIVE

Section 0 v1.0 §0.8 left a forward reference to two Cubic ingredients of the RP variational target $I(S)$. Section 1 resolved ingredient (i); Section 2 resolved ingredient (ii) and closed Open Target 1. **This merged Section 3 derives the first dynamical consequence of the operational RP:** the burden-clock suppression $\alpha(B) = 1/(1+\lambda B)$, the Effective Burden Metric Ansatz, and the Arrow of Time. The Cubic layer of the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder is now CLOSED. The Quartic layer is LIVE: the variational target $I(S)$ is fully grounded, and its dynamical consequences are being derived. The remaining Quartic-layer content — full variational solution, field equations, Einstein recovery, cosmological expansion — is deferred to Sections 5 and 8.

§3.8 Architectural Summary

LAYER $Q \rightarrow C \rightarrow Q'$

Source: synthesis of RCF_n §3 + Section_3_2 §3 + Construction Spec v1.0 Ch. 5–9. All patches implemented (P1–P5).

Section 3 reconstructed emergent temporal duration from the interplay of causal order and constraint burden on the full physical state. The reconstruction proceeds in eight stages, listed in Table 3.8.1 with their layer assignments, sources, epistemic status, and notes on patches and forward references.

§	Unit	Layer	Source	Status	Notes / Forward Refs
§3.0	Purpose & Two Temporal Scales	$L \rightarrow Q$	RCF_n §3.0 + Sec_3_2 §3.0	Established	SOE/MOE two-scale temporal structure
§3.1.1	Local Obstruction Burden $B(R)$	$Q \rightarrow C$	RCF_n §3.1 + Sec_3_1 + Sec_3_2	Established	P1: $B = \text{Tr}_R(\rho_\infty F)$ on full state (not sector)
§3.1.2	Burden Positivity	C	RCF_n §3.1 (Thm 3.1) + Sec_3_2	Established	$B \geq 0$; $B = 0$ iff first-class
§3.1.3	Burden Additivity	C	Sec_3_2 §3.1.4	Established	$B(R_1 \cup R_2) = B(R_1) + B(R_2) + I(R_1, R_2)$
§3.1.4	Three-Component Decomposition	C	RCF_n §3.1.5 (Def 3.2, Thm 3.2)	Established	mode + interaction + relational
§3.1.5	Burden Is Not Violation	C	RCF_n §3.1.6	Established	$\omega(M)=0$ while $B>0$ possible
§3.2.1	Raw Causal Depth $d\sigma$	C	Sec_3_2 §3.2.1	Established	$d\sigma = \varepsilon \cdot dn = (1/\gamma) \cdot dn$
§3.2.2	Clock Suppression DERIVED	C	Sec_3_2 §3.2 (Thm 3.2.1)	Strengthened	P2: $\alpha(B)$ from SOE/MOE ratio; $\lambda = \eta/\gamma$
§3.2.3	Monotonic Suppression	C	RCF_n §3.2 (Thm 3.3)	Established	$B_1 > B_2 \Rightarrow \alpha(B_1) < \alpha(B_2)$
§3.2.5	Gravitational Time Dilation	C	Sec_3_2 §3.2.2	Established	Fwd-ref: §5 mass-burden identity $m \equiv B_0$
§3.3.1	Burden-Weighted Proper Time	C	RCF_n §3.3 (Def 3.4)	Established	$\tau[\gamma] = \sum \varepsilon \cdot \alpha(B(e_k))$
§3.3.2	Basic Bounds	C	RCF_n §3.3 (Thm 3.4)	Established	$0 < \tau[\gamma] \leq (n+1) \cdot \varepsilon$
§3.3.4	Equal-Depth Comparison	C	RCF_n §3.3 (Thm 3.5)	Established	Higher $B \Rightarrow$ lower τ for equal depth

§	Unit	Layer	Source	Status	Notes / Forward Refs
§3.3.5	Velocity Time Dilation (leading-order)	C	Sec_3_2 §3.3.3	Theorem Target	Fwd-ref: §1.3.5 Lorentz factor derivation
§3.4.1	Temporal Geometry $\alpha(x)$	C	RCF_n §3.4 (Def 3.5)	Established	Spatial pattern of clock factors
§3.4.2	Non-Uniform Burden $\rightarrow$ Non-Uniform τ	C	RCF_n §3.4 (Thm 3.6)	Established	Gravitational time dilation analogue
§3.4.3	Burden-Clock Potential Φ_C	C	RCF_n §3.4 (Def 3.7)	Established	$\Phi_C = c^2_{\text{eff}} \cdot \log \alpha$
§3.4.4	Weak-Burden Expansion	C	RCF_n §3.4 (Thm 3.7)	Established	$\Phi_C \approx -c^2_{\text{eff}} \cdot \lambda \cdot B$; Fwd-ref: §5 Newtonian
§3.5.1	Burden Lapse α_B	C $\rightarrow$ Q'	RCF_n §3.5 (Def 3.7)	Established	$\alpha_B(x) = 1/(1+\lambda B(x))$
§3.5.2	Effective Temporal Line Element	C $\rightarrow$ Q'	RCF_n §3.5 (Def 3.8)	Established	$ds^2_{\text{time}} = -\alpha^2_B \cdot c^2_{\text{eff}} \cdot d\sigma^2$
§3.5.4	Effective Burden Metric Ansatz	Q'	RCF_n §3.5 + Ch. 9	Established	P3: unifies α_B (here) + h_{ij} (§2.4)
§3.5.5	Unified Time Dilation	Q'	Synthesis	Established	$d\tau = \alpha_B \cdot d\sigma \cdot \sqrt{(1-v^2/c^2)}$
§3.6.1	Irreversibility of Reconciliation	Q'	Sec_3_2 §3.4 + §0.4	Established	P4: R_t semigroup; $S(\rho)$ non-decreasing
§3.6.2	Arrow = Reconciliation Arrow	Q'	Sec_3_2 §3.4	Established	B_Δ decreases; coherence $\rightarrow$ decoherence
§3.6.3	Cosmological Initial Condition	Q'	Synthesis	Established (boundary)	Fwd-ref: §8 cosmology (one-way)
§3.7.1	Causal Reconciliation Principle (operational)	Q'	RCF_n §3.0.7 + §0.8	Established	P5: RP fully operational; Cubic closed
§3.7.2	Operational Quartic Layer	Q'	Synthesis	Established	$L \rightarrow Q \rightarrow C \rightarrow Q$ ladder: Q' now LIVE

Table 3.8.1 — Architectural summary of Section 3. 27 structural units across 3 layers ($L \rightarrow Q \rightarrow C \rightarrow Q'$). 5 patches implemented (P1–P5); 0 quarantined conjectures; 4 forward references out (all one-way): §3.2.5 $\rightarrow$ §5 (mass-burden identity), §3.3.5 $\rightarrow$ §1.3.5 (Lorentz factor, Theorem Target), §3.4.4 $\rightarrow$ §5 (Newtonian identification), §3.5.4 $\rightarrow$ §5 (full metric & curvature), §3.6.3 $\rightarrow$ §8 (cosmological initial state). Cubic layer CLOSED; Quartic layer LIVE.

The conceptual chain of this section is the strict emergence sequence: causal order $<$ (§1.1.3) + constraint burden $B(R)$ on the full physical state (§3.1, P1) $\rightarrow$ raw causal depth $d\sigma$ (§3.2.1) $\rightarrow$ clock suppression $\alpha(B)$ DERIVED from SOE/MOE ratio (§3.2, P2) $\rightarrow$ burden-weighted proper time $\tau[\gamma]$ (§3.3) $\rightarrow$ temporal geometry

$\alpha(x)$ (§3.4) → burden-clock potential Φ_C + weak-burden expansion (§3.4.3–4) → scalar burden-to-lapse bridge (§3.5.1–2) → Effective Burden Metric Ansatz unifying α_B with h_{ij} from §2.4 (§3.5.4, P3) → Arrow of Time grounded in §0.4 semigroup property (§3.6, P4) → Causal Reconciliation Principle fully operational (§3.7, P5). Each link in this chain depends only on the previous links and on the closed foundations of Sections 0–2. No link depends on a structure introduced later in the chain, and no link depends on Section 4 or beyond (except for the four documented forward references, all of which are one-way).

Section 3 — CLOSED. Pre-matter geometric substrate COMPLETE.

With Section 3 merged, the framework now possesses a complete pre-matter geometric substrate: **(1)** a 3D spatial metric $(X_\omega, \mathfrak{d}_\omega)$ from §2.4; **(2)** a 1D temporal lapse $\alpha_B(x)$ from §3.5; **(3)** a strict causal order $<$ from §1.1.3; **(4)** an operational Reconciliation Principle from §0.8 + §3.7; **(5)** an Effective Burden Metric Ansatz from §3.5.4 unifying (1) and (2); **(6)** an Arrow of Time from §3.6. The Cubic layer of the $L \rightarrow Q \rightarrow C \rightarrow Q$ emergence ladder is CLOSED; the Quartic layer is LIVE. The next task is to populate this geometric substrate with matter-like excitations — fields and particles — which is the content of Section 4.

Section 3 is now CLOSED. Section 4 (Fields and Particles) can be merged against this stable temporal foundation. Section 4 will reconstruct matter-like excitations from the relational constraint framework: fields as sections of bundles over the emergent metric substrate, particles as localised excitations of those fields, and interactions as non-additive composition costs (the interaction burden ρ_{int} of §3.1.4). The mass-burden identity $m \equiv B_0$ (forward reference from §3.2.5) will be derived in §4.2, completing the certification of Corollary 3.2.5 (Gravitational Time Dilation). Once Section 4 is merged, the framework will possess a complete matter+geometry substrate, setting the stage for the gravitational field equation and GR recovery in Section 5.